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PRACTICAL 3

String Functions in SQL*Plus (Oracle) & MySQL

String functions allow you to **manipulate and process text data** in SQL. Below is a detailed comparison of **SQL*Plus (Oracle)** and **MySQL** string functions, including examples.

1. String Functions in SQL*Plus (Oracle)

1.1 CONCAT – String Concatenation

```
SELECT CONCAT('Hello', ' World') FROM dual; -- Result: Hello World
SELECT 'Hello' || ' World' FROM dual; -- Alternative method using ||
```

```
SQL> select concat('hello', 'world')from dual;

CONCAT('HE
-----
helloworld

SQL> |
```

1.2 LENGTH – String Length

```
SELECT LENGTH('Oracle Database') FROM dual; -- Result: 16
```

```
SQL> SELECT LENGTH('Oracle Database') FROM dual;  
LENGTH('ORACLEDATABASE')  
-----  
15  
  
SQL> |
```

1.3 SUBSTR – Extract Substring

```
SQL> SELECT SUBSTR('Oracle Database', 8, 3) FROM dual;  
SUB  
---  
Dat  
  
SQL> |
```

```
SELECT SUBSTR('Oracle Database', 8, 3) FROM dual;
```

Extracts 'Dat' (Start from 8, length 3)

1.4 INSTR – Find Position of a Substring

```
SELECT INSTR('Oracle Database', 'D') FROM dual; -- Finds position of  
'D' (Result: 8)
```

```
SQL> SELECT INSTR('Oracle Database', 'D') FROM dual;  
  
INSTR('ORACLEDATABASE', 'D')  
-----  
8
```

1.5 REPLACE – Replace a Substring

```
SELECT REPLACE('Oracle Database', 'Database', 'SQL') FROM dual; --  
Result: Oracle SQL
```

```
SQL> SELECT REPLACE('Oracle Database', 'Database', 'SQL') FROM dual;  
  
REPLACE('O  
-----  
Oracle SQL
```

1.6 TRANSLATE – Replace Multiple Characters

```
SELECT TRANSLATE('123-456-7890', '123', 'XYZ') FROM dual; -- Result:
```

XYZ-456-7890

```
SQL> SELECT TRANSLATE('123-456-7890', '123', 'XYZ') FROM dual;  
  
TRANSLATE('1  
-----  
XYZ-456-7890
```

1.7 TRIM – Remove Spaces or Characters

SELECT TRIM(' Oracle ') FROM dual; -- Removes leading and trailing spaces

SELECT TRIM('O' FROM 'Oracle') FROM dual; -- Removes 'O' from both ends

```
SQL> SELECT TRIM(' Oracle ') FROM dual;  
  
TRIM('-----  
Oracle  
  
SQL> SELECT TRIM('O' FROM 'Oracle') FROM dual;  
  
TRIM(-----  
racle
```

1.8 LPAD & RPAD – Padding Strings

SELECT LPAD('Oracle', 10, '*') FROM dual; -- Result: ****Oracle

SELECT RPAD('Oracle', 10, '*') FROM dual; -- Result: Oracle****

```

SQL> SELECT LPAD('Oracle', 10, '*') FROM dual;

LPAD('ORAC
-----
****Oracle

SQL> SELECT RPAD('Oracle', 10, '*') FROM dual;

RPAD('ORAC
-----
Oracle****

```

1.9 LOWER, UPPER, INITCAP – Case Conversion

```

SELECT LOWER('Oracle Database') FROM dual; -- Result: oracle database
SELECT UPPER('Oracle Database') FROM dual; -- Result: ORACLE DATABASE
SELECT INITCAP('oracle database') FROM dual; -- Result: Oracle
Database

```

```

SQL> SELECT LOWER('Oracle Database') FROM dual;

LOWER('ORACLEDA
-----
oracle database

SQL> SELECT UPPER('Oracle Database') FROM dual;

UPPER('ORACLEDA
-----
ORACLE DATABASE

SQL> SELECT INITCAP('oracle database') FROM dual;

INITCAP('ORACLE
-----
Oracle Database

```

1.10 REGEXP Functions – Regular Expressions

SELECT REGEXP_SUBSTR('A123B456C', '[0-9]+') FROM dual; -- Extracts first number (Result: 123)

SELECT REGEXP_REPLACE('abc123xyz', '[0-9]', '*') FROM dual; -- Replaces digits with '*' (Result: abc***xyz)

```

SQL> SELECT REGEXP_SUBSTR('A123B456C', '[0-9]+') FROM dual;

REG
---
123

SQL> SELECT REGEXP_REPLACE('abc123xyz', '[0-9]', '*') FROM dual;

REGEXP_RE
-----
abc***xyz

```

2. String Functions in MySQL

2.1 CONCAT – String Concatenation

```
SELECT CONCAT('Hello', ' World'); -- Result: Hello World
```

2.2 LENGTH – String Length

```
SELECT LENGTH('MySQL Database'); -- Result: 15
```

2.3 SUBSTRING – Extract Substring

```
SELECT SUBSTRING('MySQL Database', 8, 3); -- Extracts 'Dat' (Start  
from 8, length 3)
```

2.4 LOCATE & INSTR – Find Position of a Substring

```
SELECT LOCATE('D', 'MySQL Database'); -- Result: 8  
SELECT INSTR('MySQL Database', 'D'); -- Result: 8
```

2.5 REPLACE – Replace a Substring

```
SELECT REPLACE('MySQL Database', 'Database', 'Server'); -- Result:  
MySQL Server
```

2.6 TRIM – Remove Spaces or Characters


```
SELECT TRIM(' MySQL '); -- Removes leading and trailing spaces
SELECT TRIM('M' FROM 'MySQL'); -- Removes 'M' from both ends
```

2.7 LPAD & RPAD – Padding Strings

```
SELECT LPAD('MySQL', 10, '*'); -- Result: *****MySQL
SELECT RPAD('MySQL', 10, '*'); -- Result: MySQL*****
```

2.8 LOWER, UPPER – Case Conversion

```
SELECT LOWER('MySQL Database'); -- Result: my database
SELECT UPPER('MySQL Database'); -- Result: MYSQL DATABASE
```

2.9 REGEXP Functions – Regular Expressions

```
SELECT REGEXP_SUBSTR('abc123xyz', '[0-9]+'); -- Extracts first number
(Result: 123)
SELECT REGEXP_REPLACE('abc123xyz', '[0-9]', '*'); -- Replaces digits
with '*' (Result: abc***xyz)
```

3. Key Differences Between SQL*Plus (Oracle) and MySQL String Functions

Function	Oracle (SQL*Plus)	MySQL
Concatenation	CONCAT(str1, str2) or ''	

Substring	SUBSTR(str, start, length)	SUBSTRING(str, start, length)
Find Position	INSTR(str, substring)	LOCATE(substring, str) or INSTR(str, substring)
Replace Substring	REPLACE(str, old, new)	REPLACE(str, old, new)
Trim Spaces	TRIM(str)	TRIM(str)
Padding	LPAD(str, length, pad_char), RPAD(str, length, pad_char)	LPAD(str, length, pad_char), RPAD(str, length, pad_char)
Case Conversion	UPPER(str), LOWER(str), INITCAP(str)	UPPER(str), LOWER(str)
Regular Expressions	REGEXP_SUBSTR(), REGEXP_REPLACE()	REGEXP_SUBSTR(), REGEXP_REPLACE()

4. Special Notes

- Oracle has **INITCAP()**, which capitalizes the first letter of each word, whereas MySQL does **not**.
- **CONCAT()** in Oracle only takes **two** arguments, while in MySQL it can take **multiple**.
- Regular expressions (**REGEXP_...**) are available in both, but Oracle has more advanced capabilities.